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ELECTRONIC DEVICE INCLUDING ANTENNA

Abstract

An electronic device is provided, which includes a first housing, a display unit included in the first housing, a second housing, and a hinge configured to couple the first housing and the second housing, wherein the display unit comprises a first printed circuit board (PCB), a first antenna pattern at least partially disposed at a first side surface of the first housing adjacent to the hinge, a conductive member electrically connected to the first antenna pattern, and a first conductive plate electrically connected to at least a portion of the conductive member, and wherein the first antenna pattern is coupled to at least a portion of the conductive member using a first fastening member and a second fastening member

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. Ser. No. 17/834,425, which was filed at the United States Patent Office on Jun. 7, 2022, which is a Bypass Continuation Application of International Application No.

PCT/KR2022/007988, which was filed on Jun. 7, 2022, and is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0094221, which was filed in the Korean Intellectual Property Office filed on Jul. 19, 2021, the entire disclosure of each of which is incorporated herein by reference.

BACKGROUND

1. Field

[0002] The disclosure relates generally to an electronic device including an antenna, and more particularly, to an electronic device capable of securing a ground of an antenna pattern using a conductive member and shielding noise transferred from electronic components disposed around the antenna pattern.

2. Description of Related Art

[0003] The use of electronic devices such as foldable type smart phones, notebook computers, or tablet personal computers (PCs) is increasing, and various functions are provided through the electronic devices.

[0004] For example, an electronic device may transmit and receive various data to and from another electronic device through wireless communication.

[0005] The electronic device may provide services such as a global positioning system (GPS), Wi-Fi, long-term evolution (LTE), sub6, 5th generation (5G), near field communication (NFC), Bluetooth®, and/or magnetic secure transmission (MST) communication.

[0006] An electronic device may include at least one antenna (e.g., an antenna pattern) in order to provide various services related to wireless communication. The antenna may be electrically connected to a controller (e.g., a processor or a wireless communication module) disposed on a printed circuit board (PCB), and transmit and receive wireless signals.

[0007] In the electronic device, when a housing (e.g., a side member) forming an external shape of the electronic device is made of a non-conductive material (e.g., a polymer), it may be difficult securely ground the antenna.

[0008] Further, when the antenna is disposed adjacent to a hinge part of a foldable electronic device and noise generated in a PCB of a display unit may be transferred to the antenna, which may deteriorate a radiation performance of the antenna.

[0009] When the ground of the antenna is not sufficiently secured and a radiation performance of the antenna is deteriorated, the electronic device may have difficulty in transmitting and receiving data to and from another electronic device.

SUMMARY

[0010] The disclosure has been made to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below.

[0011] An aspect of the disclosure is to provide an electronic device capable of securing a ground of an antenna pattern using a conductive member (e.g., conductive sheet or conductive plate) and

shielding noise transferred from electronic components disposed near the antenna pattern.

[0012] In accordance with an aspect of the disclosure, an electronic device is provided, which includes a display unit included in a first housing; a second housing; and a hinge part configured to couple the first housing and the second housing. The first housing includes a first PCB; a first antenna pattern at least partially disposed at a first side surface of the first housing adjacent to the hinge part; a conductive member electrically connected to the first antenna pattern; and a first conductive plate electrically connected to at least a portion of the conductive member. The second housing includes a second conductive plate electrically connected to the first conductive plate; a third conductive plate electrically connected to the second conductive plate; and a second PCB at which a ground electrically connected to the third conductive plate is disposed.

[0013] In accordance with another aspect of the disclosure, an electronic device is provided, which includes a display unit included in a first housing; a second housing; and a hinge part configured to couple the first housing and the second housing. The first housing includes a first PCB; a first antenna pattern disposed at least partially on a first side surface of the first housing adjacent to the hinge part and a second antenna pattern disposed in area spaced apart from the first antenna pattern; a conductive member electrically connected to the first antenna pattern and the second antenna pattern; and a first conductive plate electrically connected to at least a portion of the conductive member. The second housing includes a second conductive plate electrically connected to the first conductive plate; a third conductive plate electrically connected to the second conductive plate; and a second PCB at which a ground electrically connected to the third conductive plate is disposed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The above and other aspects, features, and advantages of the disclosure will be more apparent from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0015] FIG. 1 illustrates an electronic device in a network environment according to an embodiment;

[0016] FIG. 2 illustrates an electronic device according to an embodiment;

[0017] FIG. 3 illustrates the display unit of the electronic device of FIG. 2, according to an embodiment;

[0018] FIG. 4 illustrates a side surface of the portion A of the display unit illustrated in FIG. 3, according to an embodiment;

[0019] FIG. 5 illustrates a conductive member and a signal connection member according to an embodiment;

[0020] FIG. 6 illustrates a first antenna pattern, a conductive member, and a first PCB of an electronic device according to an embodiment; and

[0021] FIG. 7 illustrates an internal configuration of the body part of the electronic device of FIG. 2, according to an embodiment.

DETAILED DESCRIPTION

[0022] Hereinafter, various embodiments will be described in detail with reference to attached drawings. In the disclosure, embodiments are described in the drawings and a related detailed description is set forth, but this is not intended to limit the embodiments of the disclosure. Descriptions of well-known functions and constructions are omitted for the sake of clarity and conciseness.

[0023] FIG. 1 illustrates an electronic device in a network environment according to an embodiment.

[0024] Referring to FIG. 1, an electronic device **101** in a network environment **100** may

communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**). [0025] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0026] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an ISP or a CP) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the NPU) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent DNN (BRDNN), deep Q-network, or a combination of two or more thereof, but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0027] The memory **130** may store various data used by at least one component (e.g., the processor

120 or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**. [0028] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0029] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0030] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0031] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0032] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0033] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0034] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0035] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0036] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0037] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, ISPs, or flashes.

[0038] The power management module **188** may manage power supplied to the electronic device

101. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0039] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0040] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more CPs that are operable independently from the processor **120** (e.g., the AP) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth®, wireless-fidelity (Wi-Fi) direct, or IR data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the SIM **196**.

[0041] The wireless communication module **192** may support a 5G network, after a 4th generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0042] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a PCB). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**)

from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**

[0043] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a PCB, an RFIC disposed on a first surface (e.g., the bottom surface) of the PCB, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the PCB, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0044] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0045] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or MEC. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0046] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0047] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment.

[0048] With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly

indicates otherwise.

[0049] As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order).

[0050] It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0051] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0052] FIG. 2 illustrates an electronic device according to an embodiment.

[0053] Referring to FIG. 2, an electronic device **201** is a notebook computer. However, the disclosure is not limited thereto, and may be applied to various types of electronic devices including a hinge part such as a foldable type smart phone or a tablet PC.

[0054] The electronic device **201** includes a display unit **210**, a hinge part **215**, and a body part **220**.

[0055] The display unit **210** includes a display **212**, a first housing **230**, and a camera module **214**.

[0056] The display **212** may display various menus of the electronic device **201**, information input by a user using a keyboard **202**, and/or information to be provided to the user. The display **212** may include at least one of a liquid crystal display (LCD), an organic light emitted diode (OLED), an active matrix organic light emitted diode (AMOLED), a flexible display, or a transparent display. The display **212** may provide various screens, such as a home screen, a menu screen, a lock screen, a game screen, a web page screen, a call screen, or a music or video playback screen, etc., according to the use of the electronic device **201**.

[0057] The first housing **230** forms an external shape of the display unit **210**. The first housing **230** may be made of a conductive material (e.g., metal) and/or a non-conductive material (e.g., a polymer). The first housing **230** may protect the display **212** and/or the camera module **214** included in the display unit **210**. The first housing **230** may protect a first PCB and other electronic components included inside the display unit **210**.

[0058] The first housing **230** includes a first side surface **231** at a surface adjacent to a hinge part **215**. The first side surface **231** may at least partially include a conductive material (e.g., metal).

[0059] The camera module **214** may capture a still picture and/or a moving picture. The camera module **214** may include one or more lenses, image sensors, ISPs, or flashes.

[0060] The hinge part **215** may couple the display unit **210** and a body part **220** to be foldable and/or unfoldable. The hinge part **215** may include a hinge module that mechanically connects the display unit **210** and the body part **220**. The hinge part **215** may include at least a portion of at least one signal connection member (e.g., a flexible PCB (FPCB) or coaxial cable) that electrically connects the display unit **210** and the body part **220** therein. At least one hinge part **215** may be included in the electronic device **201** so as to couple the display unit **210** and the body part **220** in a foldable fashion.

[0061] The body part **220** includes a keyboard **202**, a touch pad **204**, a palm rest **206**, and a second housing **240**. The body part **220** may also include a display.

[0062] The keyboard **202** may have a plurality of keys at an upper portion (e.g., in the z-axis direction) of the body part **220**. The keyboard **202** may include a plurality of input keys and

function keys for configuring various functions of the electronic device **201**. The function keys may include a direction key, a volume key, and a shortcut key configured to perform a specific function. The keyboard **202** may include any one of a query keypad, a 3*4 keypad, a 4*3 keypad, or a touch key.

[0063] The touch pad **204** may replace a function of a mouse. The touch pad **204** may be used for inputting a command for selecting or executing various screens and/or applications displayed through the display unit **210**.

[0064] The palm rest **206** may support a user hand to reduce wrist fatigue when using the keyboard **202**.

[0065] The second housing **240** forms an external shape of the body part **220**. The second housing **240** may be made of a conductive material (e.g., metal) and/or a non-conductive material (e.g., a polymer). The second housing **240** includes the palm rest **206** formed in the body part **220**. The second housing **240** may protect a second PCB and/or a battery included inside the body part **220**. In the second PCB, various electronic components may be disposed.

[0066] The second housing **240** includes a first surface **241** (e.g., an upper surface) disposed in a first direction (e.g., z-axis direction) of the body part **220**, a second surface **245** (e.g., a lower surface) disposed in a second direction (e.g., -z-axis direction), which is a direction opposite to that of the first surface **241**, and a side surface **243** enclosing a space between the first surface **241** and the second surface **245**.

[0067] The second housing **240** includes a second side surface **242** at a surface adjacent to the hinge part **215**.

[0068] FIG. **3** illustrates the display unit of the electronic device of FIG. **2**, according to an embodiment. Specifically, FIG. **3** illustrates the display unit **210** of the electronic device **201** viewed from one direction (e.g., y-axis direction).

[0069] FIG. **4** illustrates a side surface of the portion A of the display unit illustrated in FIG. **3**, according to an embodiment.

[0070] Referring to FIGS. **3** and **4**, the display unit **210** of the electronic device **201** includes a first antenna pattern **301**, a second antenna pattern **303**, a conductive member **310**, a first conductive plate **320**, and a first PCB **330**.

[0071] The first antenna pattern **301** may be at least partially disposed in the first housing **230** forming an external shape of the display unit **210**. For example, the first antenna pattern **301** may be at least partially disposed at the first side surface **231** of the first housing **230** adjacent to the hinge part **215**.

[0072] The first antenna pattern **301** may be electrically connected to a processor and/or a wireless communication module, and may perform a function of the first antenna. The first antenna pattern **301** and/or the second antenna pattern **303** may operate in substantially the same or different frequency bands. For example, the first antenna pattern **301** may operate in a first WiFi frequency band (e.g., about 2.4 Ghz-2.5 Ghz or about 5 Ghz-6 Ghz), a GPS frequency band (e.g., about 1.5 Ghz-1.6 Ghz) and/or a Bluetooth® frequency band (e.g., about 2.4 Ghz-2.5 Ghz).

[0073] The second antenna pattern **303** may be at least partially disposed at the first side surface **231** of the first housing **230** adjacent to the hinge part **215**. The second antenna pattern **303** may be disposed to be spaced apart from the first antenna pattern **301**. The second antenna pattern **303** may be disposed in a first direction (e.g., x-axis direction) based on the first antenna pattern **301**.

[0074] The second antenna pattern **303** may be electrically connected to a processor and/or a wireless communication module, and may perform a function of the second antenna. The second antenna pattern **303** may operate in a second WiFi frequency band (e.g., about 2.4 Ghz-2.5 Ghz or about 5 Ghz-6 Ghz), a GPS frequency band (e.g., about 1.5 Ghz-1.6 Ghz) and/or a Bluetooth® frequency band (e.g., about 2.4 Ghz-2.5 Ghz).

[0075] The first antenna pattern **301** may operate in a first WiFi frequency band (e.g., about 2.4 Ghz-2.5 Ghz), and the second antenna pattern **303** may operate in a second WiFi frequency band

(e.g., about 5 GHz-6 GHz), or the first antenna pattern **301** may operate in the second WiFi frequency band, and the second antenna pattern **303** may operate in the first WiFi frequency band. [0076] The first antenna pattern **301** and the second antenna pattern **303** may operate in the above-described first and second Wi-Fi frequency bands, respectively (e.g., about 2.4 GHz-2.5 GHz or about 5 GHz-6 GHz), a GPS frequency band (e.g., about 1.5 GHz-1.6 GHz) and/or a Bluetooth® frequency band (e.g., about 2.4 GHz-2.5 GHz).

[0077] The frequency bands of each of the first antenna pattern **301** and the second antenna pattern **303** are not limited to the above-described examples, and the first antenna pattern **301** and the second antenna pattern **303** may operate in other various frequency bands (e.g., sub 6 and 5G).

[0078] Although the first antenna pattern **301** and the second antenna pattern **303** are illustrated in the embodiments of FIGS. 3 and 4, the electronic device **201** may alternatively include only one of the first antenna pattern **301** or the second antenna pattern **303**, or more than two antenna patterns.

[0079] The first antenna pattern **301** and/or the second antenna pattern **303** may be electrically connected to a second PCB disposed in the body part **220** using a signal connection member **340** (e.g., a coaxial cable). The second PCB may include a processor and/or a wireless communication module.

[0080] As another example, when a processor and/or a wireless communication module are/is disposed on the first PCB **330**, the signal connection member **340** may be electrically connected to the first PCB **330**.

[0081] The signal connection member **340** may include a first signal connection member and a second signal connection member.

[0082] The conductive member **310** may be electrically connected to the first antenna pattern **301** and/or the second antenna pattern **303**. The conductive member **310** may include a conductive sheet or a conductive plate. At least a portion of the conductive member **310** may be electrically connected to the first antenna pattern **301**. At least a portion of the conductive member **310** may be electrically connected to the second antenna pattern **303**.

[0083] The first conductive plate **320** may be electrically connected to at least a portion of the conductive member **310**. For example, the first conductive plate **320** may be electrically connected to the conductive member **310** using a contact part **315**. The first conductive plate **320** may be disposed in a second direction (e.g., -x-axis direction) based on the first antenna pattern **301**.

[0084] The first conductive plate **320** may be electrically connected to a second conductive plate and a third conductive plate disposed in the body part **220**. The first conductive plate **320** may be coupled to at least a portion of the first housing **230** or the first side surface **231** of the display unit **210** using at least one screw **325**.

[0085] The first conductive plate **320** may be electrically connected to the second conductive plate and the third conductive plate disposed in the body part **220**. For example, the first conductive plate **320**, the second conductive plate, and the third conductive plate may include stainless use steel (SUS).

[0086] The first PCB **330** may be disposed under the conductive member **310** (e.g., -z-axis direction). For example, the first PCB **330** may be disposed inside the first housing **230** rather than the conductive member **310**. The first PCB **330** may include at least one electronic component and/or a connector for driving the display **212**.

[0087] The conductive member **310** may be disposed between the first antenna pattern **301** and/or the second antenna pattern **303** and the first PCB **330**. The first antenna pattern **301** and/or the second antenna pattern **303** may be positioned in a -y-axis direction of the conductive member **310**, and the first PCB **330** may be positioned in a -z-axis direction of the conductive member **310**. The conductive member **310** may be electrically connected (e.g., connected to the ground) to the first antenna pattern **301** and/or the second antenna pattern **303** and the first PCB **330**. The conductive member **310** may shield noise and/or electromagnetic interference generated in the first PCB **330** from being transferred to the first antenna pattern **301** and/or the second antenna pattern

303.

[0088] FIG. 5 illustrates a conductive member and a signal connection member according to an embodiment.

[0089] Referring to FIG. 5, the conductive member **310** may be coupled to the first antenna pattern **301** and/or the second antenna pattern **303** using a first fastening member **501**, a second fastening member **503**, a third fastening member **505**, and/or a fourth fastening member **507**.

[0090] The first fastening member **501**, the second fastening member **503**, the third fastening member **505**, and the fourth fastening member **507** may be each coupled to the conductive member **310**, and be spaced apart. The first fastening member **501**, the second fastening member **503**, the third fastening member **505**, and/or the fourth fastening member **507** may be made of a non-conductive injection molding material. The first fastening member **501**, the second fastening member **503**, the third fastening member **505**, and/or the fourth fastening member **507** may be made of a non-conductive material (e.g., polymer). The first fastening member **501**, the second fastening member **503**, the third fastening member **505**, and/or the fourth fastening member **507** may be attached to a support member (e.g., antenna carrier) in which the first antenna pattern **301** and/or the second antenna pattern **303** are/is formed and couple the conductive member **310** to the first antenna pattern **301** and/or the second antenna pattern **303**.

[0091] The first antenna pattern **301** may be interposed between the first fastening member **501** and the second fastening member **503**. The first antenna pattern **301** may be fixedly supported by the first fastening member **501** and the second fastening member **503**. At least a portion of the first antenna pattern **301** may be electrically connected to the conductive member **310** using, e.g., a conductive tape. The first antenna pattern **301** may be electrically connected to at least a portion of the conductive member **310** using the first fastening member **501** and the second fastening member **503**. The second antenna pattern **303** may be interposed between the third fastening member **505** and the fourth fastening member **507**. The second antenna pattern **303** may be fixedly supported by the third fastening member **505** and/or the fourth fastening member **507**. At least a portion of the second antenna pattern **303** may be electrically connected to the conductive member **310** using, e.g., a conductive tape. The second antenna pattern **303** may be electrically connected to at least a portion of the conductive member **310** using the third fastening member **505** and the fourth fastening member **507**.

[0092] The first antenna pattern **301** and the second antenna pattern **303** may be electrically connected to a second PCB disposed in the body part **220** using the signal connection member **340** (e.g., a coaxial cable). The signal connection member **340** includes a first signal connection member **340a** and a second signal connection member **340b**. The first signal connection member **340a** may include a first feeding line. The second signal connection member **340b** may include a second feeding line. A first end **341** of the first signal connection member **340a** may be electrically connected to the first antenna pattern **301**. A second end **342** of the first signal connection member **340a** may be electrically connected to the second PCB disposed in the body part **220**. A first end **345** of the second signal connection member **340b** may be electrically connected to the second antenna pattern **303**. A second end **346** of the second signal connection member **340b** may be electrically connected to the second PCB disposed in the body part **220**.

[0093] FIG. 6 illustrates a first antenna pattern, a conductive member, and a first PCB of an electronic device according to an embodiment.

[0094] Referring to FIG. 6, a conductive member **310** is interposed between a first antenna pattern **301** (or the second antenna pattern **303**) and a first PCB **330**. The conductive member **310** may be electrically connected to the first antenna pattern **301** and the first PCB **330**.

[0095] The first PCB **330** includes at least one electronic component **620** and a connector **610** for driving the display **212**. The connector **610** and the at least one electronic component **620** may transfer noise to the first antenna pattern **301** or may cause electromagnetic interference. The at least one electronic component **620** may include a PMIC and/or a drive integrated circuit (IC).

[0096] The conductive member **310** may shield noise and/or electromagnetic interference transferred from the first PCB **330** to the first antenna pattern **301**. By shielding noise and/or electromagnetic interference transferred from the first PCB **330** to the first antenna pattern **301** using the conductive member **310**, a radiation performance of the first antenna pattern **301** may be reduced from being deteriorated.

[0097] Alternatively, the conductive member **310** may be disposed in the support member **605** (e.g., antenna carrier). For example, the first fastening member **501**, the second fastening member **503**, the third fastening member **505**, and/or the fourth fastening member **507** as illustrated in FIG. 5 may be coupled to the support member **605**.

[0098] FIG. 7 illustrates an internal configuration of the body part of the electronic device of FIG. 2 according to an embodiment. Specifically, FIG. 7 illustrates the body part **220** of the electronic device **201** in FIG. 2 viewed from one direction (e.g., the z-axis direction).

[0099] Referring to FIG. 7, the body part **220** includes a second conductive plate **350**, a third conductive plate **710**, and a second PCB **720**.

[0100] The second conductive plate **350** may be electrically connected to the first conductive plate **320** as illustrated in FIG. 3.

[0101] The third conductive plate **710** may be electrically connected to the second conductive plate **350**, e.g., using at least one screw **325**.

[0102] The third conductive plate **710** may be electrically connected to a ground **730** formed in the second PCB **720**. For example, the ground **730** may be disposed in a layer of the second PCB **720**, and a part thereof may be open to an outermost portion of the second PCB **720** to be electrically connected to the third conductive plate **710**. The ground **730** may be formed using a conductive metal (e.g., copper foil).

[0103] The second PCB **720** may include a processor, a memory, a power management module, a communication module, an interface and/or a connection terminal for driving the electronic device **201**. The processor and/or the communication module disposed in the second PCB **720** may be electrically connected to the first antenna pattern **301** and/or the second antenna pattern **303** disposed in the display unit **210** using the signal connection member **340** (e.g., coaxial cable).

[0104] Referring to FIGS. 3 and 7, the first antenna pattern **301** and/or the second antenna pattern **303** disposed at least partially at the first side surface **231** of the display unit **210** may be electrically connected to the first conductive plate **320** using the conductive member **310**. The first conductive plate **320** may be electrically connected to the second conductive plate **350** and the third conductive plate **710** at least partially disposed adjacent to the second side surface **242** of the body part **220**. The third conductive plate **710** may be electrically connected to the ground **730** disposed in the second PCB **720**. Using the above-described configuration, the first antenna pattern **301** and the second antenna pattern **303** may secure a ground.

[0105] An electronic device according to an embodiment of the disclosure may include a display unit including a first housing; a body part including a second housing; and a hinge part configured to couple the display unit and the body part to be foldable and/or unfoldable, wherein the display unit may include a first PCB; a first antenna **301** at least partially disposed at a first side surface of the first housing adjacent to the hinge part; a conductive member electrically connected to the first antenna pattern; and a first conductive plate electrically connected to at least a portion of the conductive member, wherein the body part may include a second conductive plate **350** electrically connected to the first conductive plate; a third conductive plate electrically connected to the second conductive plate, and a second PCB at which a ground electrically connected to the third conductive plate is disposed.

[0106] The electronic device may further include a second antenna pattern spaced apart from the first antenna pattern at the first side surface of the first housing, wherein the conductive member may be configured to be disposed between the second antenna pattern and the first PCB **330**.

[0107] The first antenna pattern and/or the second antenna pattern may be configured to be

electrically connected to the second PCB using a signal connection member.

[0108] The first antenna pattern and the second antenna pattern may be configured to operate in different frequency bands or substantially the same frequency band.

[0109] The first antenna pattern may be configured to operate in at least one frequency band of 2.4 Ghz-2.5 Ghz, 5 Ghz-6 Ghz, or 1.5 Ghz-1.6 Ghz, and the second antenna pattern may be configured to operate in at least one frequency band of 2.4 Ghz-2.5 Ghz, 5 Ghz-6 Ghz, or 1.5 Ghz-1.6 Ghz.

[0110] The conductive member may be disposed between the first antenna pattern and the first PCB.

[0111] The conductive member may be electrically connected to the first conductive plate using a contact part.

[0112] The second antenna pattern may be disposed spaced apart in a first direction of the first antenna pattern, and the first conductive plate may be disposed in a second direction of the first antenna pattern.

[0113] The conductive member may be configured to shield noise and/or electromagnetic interference generated in the first PCB from being transferred to the first antenna pattern.

[0114] The first antenna pattern may be configured to be coupled to at least a portion of the conductive member using a first fastening member and a second fastening member, and the second antenna pattern may be configured to be coupled to at least a portion of the conductive member using a third fastening member and a fourth fastening member.

[0115] According to an embodiment of the disclosure, an electronic device may include a display unit including a first housing; a body part including a second housing; and a hinge part configured to couple the display unit and the body part to be foldable and/or unfoldable, wherein the display unit may include a first PCB; a first antenna pattern disposed at least partially at a first side surface of the first housing adjacent to the hinge part and a second antenna pattern disposed spaced apart from the first antenna pattern; a conductive member electrically connected to the first antenna pattern and the second antenna pattern; and a first conductive plate electrically connected to at least a portion of the conductive member, wherein the body part may include a second conductive plate electrically connected to the first conductive plate; a third conductive plate electrically connected to the second conductive plate; and a second PCB in which a ground electrically connected to the third conductive plate is disposed.

[0116] The first antenna pattern and the second antenna pattern may be configured to be electrically connected to the second PCB using a signal connection member.

[0117] The first antenna pattern and the second antenna pattern may be constituted to operate in different frequency bands or in substantially the same frequency band.

[0118] The first antenna pattern may be configured to operate in at least one frequency band of 2.4 Ghz-2.5 Ghz, 5 Ghz-6 Ghz, or 1.5 Ghz-1.6 Ghz, and the second antenna pattern may be configured to operate in at least one frequency band of 2.4 Ghz-2.5 Ghz, 5 Ghz-6 Ghz, or 1.5 Ghz-1.6 Ghz.

[0119] The conductive member may be disposed between the first antenna pattern and the second antenna pattern, and the first PCB.

[0120] The conductive member may be electrically connected to the first conductive plate using a contact part.

[0121] The second antenna pattern may be disposed in a first direction of the first antenna pattern, and the first conductive plate may be disposed in a second direction of the first antenna pattern.

[0122] The conductive member may be constituted to shield noise and/or electromagnetic interference generated in the first PCB from being transferred to the first antenna pattern and the second antenna pattern.

[0123] The first antenna pattern may be coupled to at least a portion of the conductive member using a first fastening member and a second fastening member, and the second antenna pattern may be configured to be coupled to at least a portion of the conductive member using a third fastening member and a fourth fastening member.

[0124] The second conductive plate may be configured to be electrically connected to the third conductive plate using at least one screw.

[0125] According to the above-described embodiments, by securing a ground of an antenna pattern (e.g., conductive pattern) using a conductive member (e.g., conductive sheet or conductive plate), and shielding noise transferred from an electronic component (e.g., a PCB of the display unit) disposed around the antenna pattern, it is possible to reduce deterioration of a radiation performance of the antenna.

[0126] Although the disclosure has been described above according to various embodiments thereof, changes and modifications made by those of ordinary skill in the art to which the disclosure pertains without departing from the technical spirit of the disclosure also belong to the disclosure.

[0127] While the disclosure has been described with reference to various embodiments, various changes may be made without departing from the spirit and the scope of the disclosure, which is defined, not by the detailed description and embodiments, but by the appended claims and their equivalents.

Claims

1. An electronic device, comprising: a first housing; a display unit included in the first housing; a second housing; and a hinge configured to couple the first housing and the second housing, wherein the display unit comprises: a first printed circuit board (PCB); a first antenna pattern at least partially disposed at a first side surface of the first housing adjacent to the hinge; a conductive member electrically connected to the first antenna pattern; and a first conductive plate electrically connected to at least a portion of the conductive member, and wherein the first antenna pattern is coupled to at least a portion of the conductive member using a first fastening member and a second fastening member.
2. The electronic device of claim 1, wherein the second housing comprises: a second conductive plate electrically connected to the first conductive plate; a third conductive plate electrically connected to the second conductive plate; and a second PCB at which a ground electrically connected to the third conductive plate is disposed.
3. The electronic device of claim 2, further comprising a second antenna pattern disposed in an area spaced apart from the first antenna pattern at the first side surface of the first housing.
4. The electronic device of claim 3, wherein the conductive member is disposed between the second antenna pattern and the first PCB.
5. The electronic device of claim 3, wherein the second antenna pattern is coupled to at least a portion of the conductive member using a third fastening member and a fourth fastening member.
6. The electronic device of claim 3, wherein at least one of the first antenna pattern or the second antenna pattern is electrically connected to the second PCB using a signal connecting member.
7. The electronic device of claim 3, wherein the first antenna pattern and the second antenna pattern operate in at least one frequency band range of 2.4 gigahertz (Ghz)-2.5 Ghz, 5 Ghz-6 Ghz, or 1.5 Ghz-1.6 Ghz.
8. The electronic device of claim 3, wherein the second antenna pattern is disposed in an area spaced apart in a first direction from the first antenna pattern.
9. The electronic device of claim 8, wherein the first conductive plate is disposed in a second direction opposite to the first direction of the first antenna pattern.
10. The electronic device of claim 1, wherein the conductive member is disposed between the first antenna pattern and the first PCB.
11. The electronic device of claim 1, wherein the conductive member is electrically connected to the first conductive plate using a contact part.
12. The electronic device of claim 1, wherein the conductive member is configured to shield at

least one of noise or electromagnetic interference generated in the first PCB from the first antenna pattern.

13. The electronic device of claim 3, wherein the conductive member is disposed between the first antenna pattern, the second antenna pattern, and the first PCB.

14. The electronic device of claim 2, wherein the second conductive plate is electrically connected to the third conductive plate using at least one screw.
